ClaimLens

AI for car insurance claim processing

Solution review · November 2026

OVERVIEW

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THE BUSINESS CHALLENGE

Filing and processing a claim is slow

Car damage claim processing can take several days. The cost falls on both parties: the car owner waits, and the insurer spends skilled time on manual review.

Meet Bo



Bo

Car owner

Bo owns a car. One afternoon, he’s in an accident.

He wants to file a claim — but first he needs photos, a repair quote from a shop, forms, and phone calls.

Meet Mindy



Mindy

Claim processor

Mindy reviews every claim by hand.

She reads the submitted quote, checks the damage against the photos, and decides whether the numbers hold up — claim after claim.

THE BUSINESS CHALLENGE

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7 days

Car owner gets a quote and submits a claim

20 claims

a processor reviews manually per day

100% manual

every quote read and checked by hand

THE SOLUTION

Meet ClaimLens

ClaimLens turns a photo of the damage into an itemized repair estimate in minutes — and routes it to a processor

FOR BO

Minutes, not days

Upload a photo, get an instant estimate, claim filed — no forms, no phone calls.

FOR MINDY

Review, don’t retype

The analysis is done for her; she confirms line items instead of building them.

FOR THE INSURER

Consistent & scalable

Every claim scored the same way; low-confidence cases flagged for review.

THE SOLUTION

How ClaimLens works

1

Upload

Bo drops in a photo of the damage from any device.

2

See the car

One vision model (GPT-4o) reads the vehicle color, model and every visible damage area.

3

Estimate cost

Each damage item is priced by severity and location into an itemized repair estimate.

4

Flag or file

Low-confidence claims are routed to a processor; the rest are filed automatically.

5

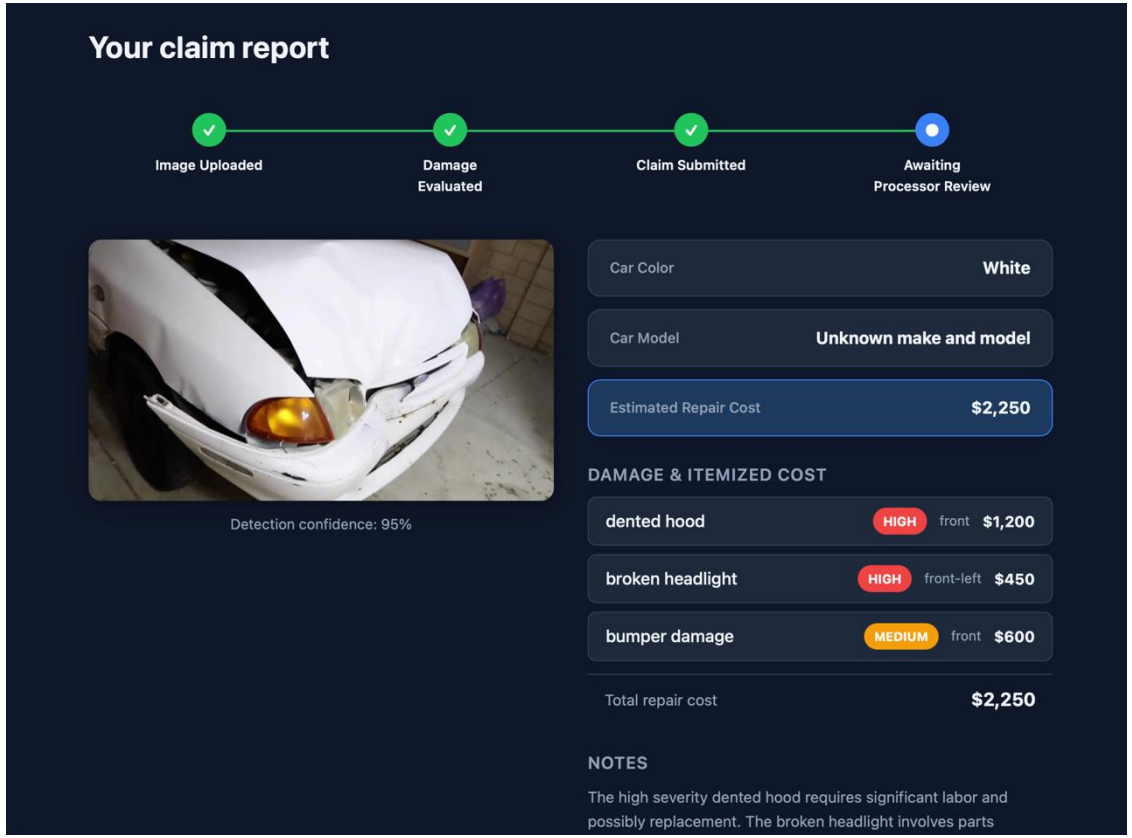
Human review

Mindy approves, declines or edits line items — then approves the claim.

THE PRODUCT

Bo can now submit claim in minutes



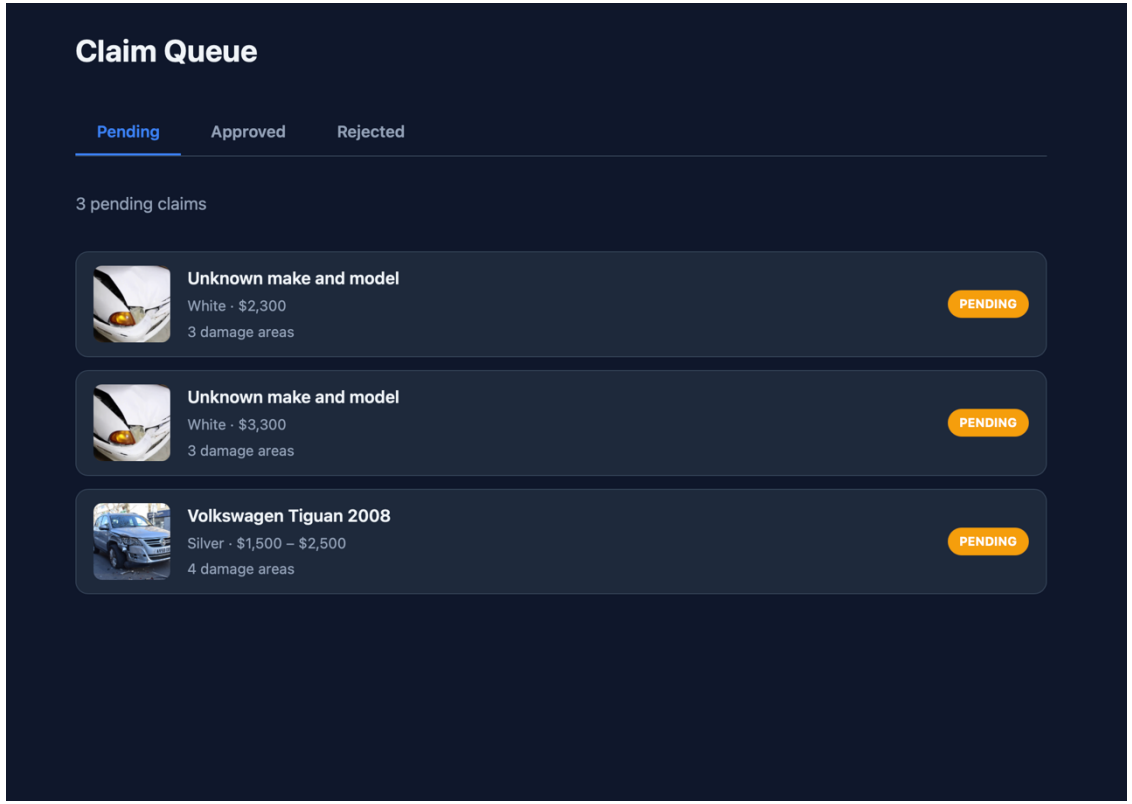


Instant claim report — itemized, with confidence

THE PRODUCT

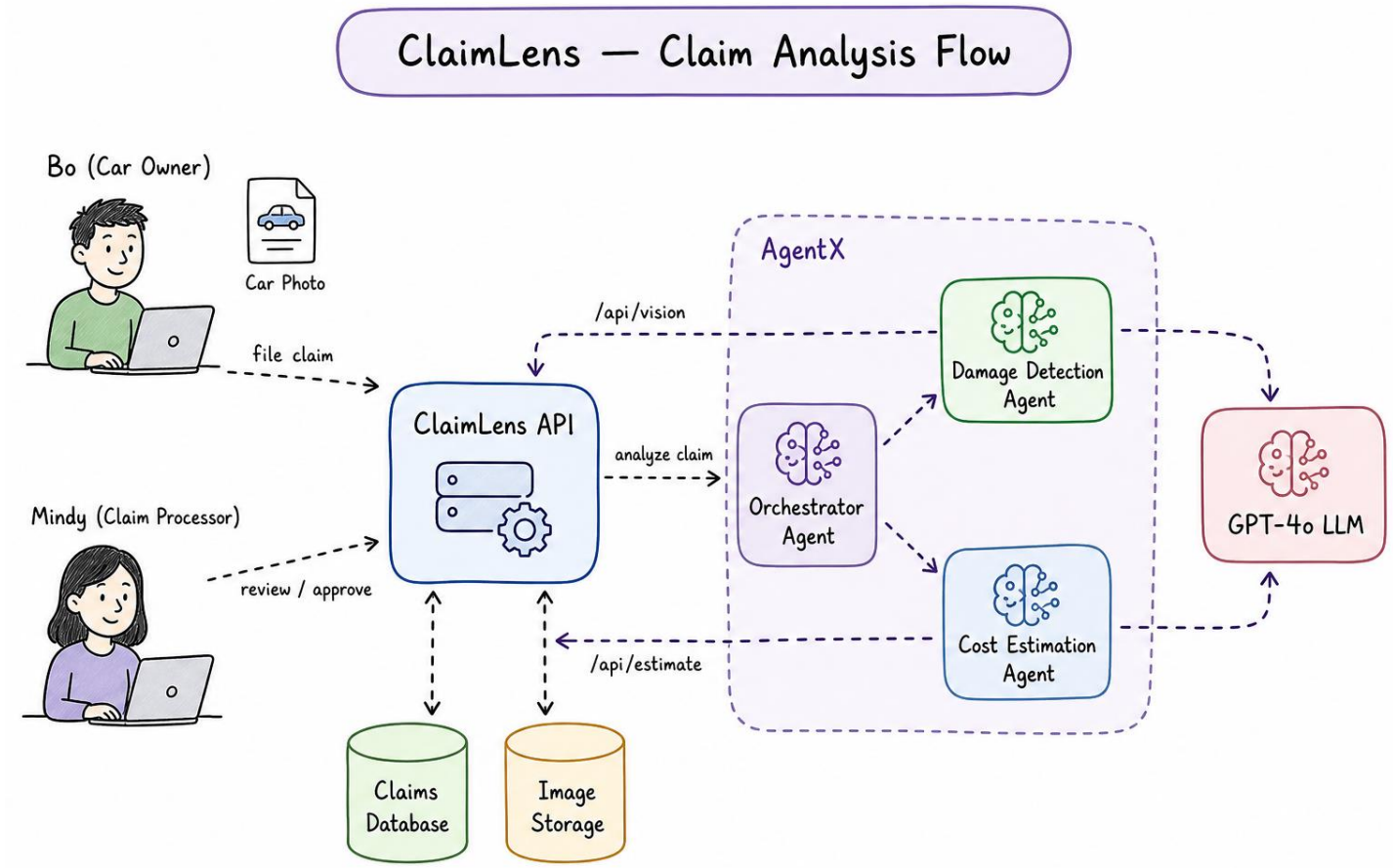
Mindy can process 5x more claims





HOW IT WORKS

Architecture



THE STACK

Frontend
React + Vite web app

API
FastAPI (Python)

Vision + cost
AgentX, GPT-4o

Database
PostgreSQL

Storage
Local / S3 for images

DOES IT WORK?

Evaluation of Car Damage Detection

25 images from the public CarDD car-damage dataset, run through the real ClaimLens pipeline (GPT-4o).

100%
detects that damage exists

0.72
damage-type F1 (micro)

0.79
precision

0.67
recall

STRONG

- Glass shatter detected perfectly (F1 1.00)
- Strong on scratch, flat tire (0.80) and dents (0.74)
- Never misses that a car is damaged

TO IMPROVE

- “Crack” not labeled — 0 recall (vocabulary gap)
- Small n = 25; no clean-car negatives yet
- Keep a human in the loop for decisions

ECONOMICS

Evaluation of Repair Cost Estimation

Each claim runs two GPT-4o calls — one to read the vehicle and damage, one to price it. We measured the repair estimate against actual repair costs on 52 damage items.

estimated repair cost vs. actual repair cost

0.90	$123	19%	75%
correlation with actual cost	mean error per damage item	error on the full claim total	items within 25% of actual

** Need to build a better dataset

ECONOMICS

Cost of AI Claim Processing

Each claim runs two GPT-4o calls — one to read the vehicle and damage, one to price it. We measured the repair estimate against actual repair costs on 52 damage items.

Cost component	Assumed usage for 100,000 claims	Total cost	Cost per claim
GPT-4o vision analysis	~1,800 input + 400 output tokens per claim	$850	$0.0085
GPT-4o cost estimation	~600 input + 300 output tokens per claim	$450	$0.0045
Cloud infrastructure	API hosting, database, image storage, networking, logging	$240	$0.0024
Total estimated cost	**100,000 claims over approximately one month**	**$1,540**	**$0.0154 · 1.54¢**

WHAT'S NEXT

Run a pilot

THE PILOT

- Shadow mode on a slice of live claims
- Human-in-the-loop on every decision
- 4 weeks, one claim category
- Compare against current manual baseline

SUCCESS CRITERIA

- Lower time-to-estimate for the customer
- More claims per processor per day
- Estimate accuracy within agreed tolerance
- Processor trust & override rate

THE RIGHT TEAM

- Claims operations lead (owns the baseline)
- Experienced claim processors as reviewers
- ML / data support for tuning & eval
- Compliance & risk sign-off

Thank you · Questions